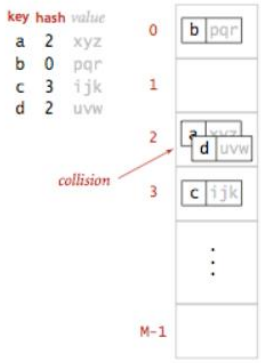
Java 12 - Hash Tables - Detyra të zgjidhura

Lënda: Hyrje në Algoritme



Hashing: the crux of the problem

Hash Tables implementojnë **Symbol Tables.**
Insertimi i elementeve behet me **O(1)**

- **Open Addressing/Closed Hashing:** linear probing, quadratic probing, double hashing, cuckoo hashing
- **Closed Addressing/Open Hashing:** separate chaining

Open Addressing: në pamundësi të vendosjes së 2 çelësave në të njejtën hapësirë adresuese (kur ka kolizion), kërkohet një hapësirë e lirë brenda hash tabelës ekzistuese
Closed Hashing: nuk lejohet të vendosen dy çelësa në të njejtin index në tabelë, konsiderohet si collision.

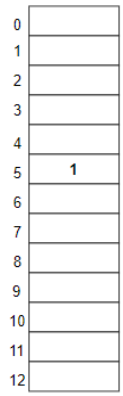
Closed Addressing: në rast të kolizionit, hapësira adresuese lidhet në linked list dhe në këtë formë 2 apo më shumë çelësa mund të jenë pjesë e të njejtës (adresë në hash tabelë)
Open Hashing: lejohet të vendosen disa çelësa në të njejtin index në tabelë, ku çelësat lidhen me anë listave të lidhura.
Linear Probing

Formula: (f(x) + i) mod M; M>N, numër i thjeshtë i=0, 1, 2…

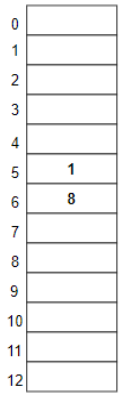
Detyra 1: Të bëhet hash vargu i dhënë: A={1, 8, 49, 18, 24, 34, 48, 5} nëse është dhënë f(x)=2x+3

N = 8, M = 13

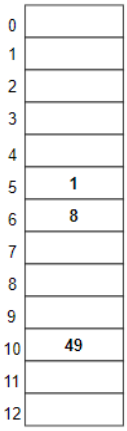
X_1 = 1, f(1) = 2*1+3=5.
h(1) = (5 + 0) mod 13
h(1)=5 mod 13
h(1)=**5**



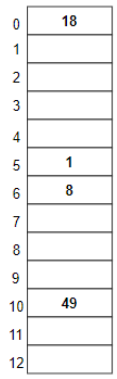
X_2 = 8, f(8) = 2*8+3=19.
h(8)=(19 + 0) mod 13
h(8)=19 mod 13
h(8) = **6**



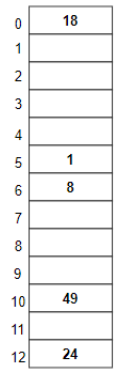
X_3 = 49, f(49) = 2*49+3 = 101.
h(49)=(101 + 0) mod 13
h(49)=101 mod 13
h(49) = **10**



X_4 = 18, f(18) = 2*18+3 = 39
h(18)=(39 + 0) mod 13
h(18)=39 mod 13
h(18)=**0**

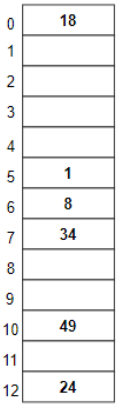


X_5 = 24, f(24) = 2*24+3=51
h(24)=(51 + 0) mod 13
h(18)=51 mod 13
h(18)=**12**



X_6 = 34, f(34) = 2*34+3=71
h(34) = (71 + 0) mod 13
h(34) = 71 mod 13
h(34) = **6 // collision, i++**

h1(34) = (71 + 1) mod 13
h1(34) = 72 mod 13
h1(34) = **7**



X_7 = 48, f(48) = 2*48+3 = 99
h(48) = (99 + 0) mod 13
h(48) = 99 mod 13
h(48) = **8**

0	18
1	
2	
3	
4	
5	1
6	8
7	34
8	48
9	
10	49
11	
12	24

X_8 = 5, f(5) = 2*5+3 = 13
h(5) = (13 + 0) mod 13
h(5) = 13 mod 13
h(5) = **0 // collision, i++**

$h_1(5)$ = (13 + 1) mod 13
$h_1(5)$ = 14 mod 13
$h_1(5)$ = **1**

0	18
1	5
2	
3	
4	
5	1
6	8
7	34
8	48
9	
10	49
11	
12	24

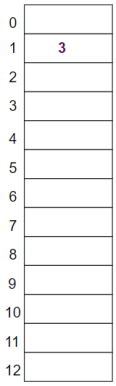
Quadratic Probing

Formula: $(f(x) + i + i^2) \bmod M$; M>N, numër i thjeshtë
i=0, 1, 2…

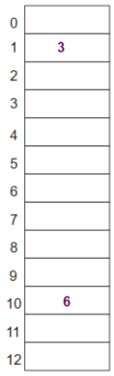
Detyra 1: Të bëhet hash vargu i dhënë: A={3, 6, 13, 24, 18, 2, 5} nëse është dhënë f(x)=3x+5

N = 7, M = 13

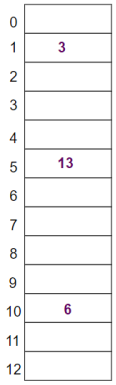
X_1 = 3, f(3) = 3*3+5=14.
h(3) = $(14 + 0 + 0^2) \bmod 13$
h(3)=14 mod 13
h(3)=**1**



X_2 = 6, f(6) = 3*6+5=23.
h(6) = $(23 + 0 + 0^2) \bmod 13$
h(6)=23 mod 13
h(6)=**10**



X_3 = 13, f(13) = 3*13+5=44.
h(13) = $(44 + 0 + 0^2) \bmod 13$
h(13)=44 mod 13
h(13)=**5**



X_4 = 24, f(24) = 3*24+5=77.
h(24) = (77 + 0 + 0^2) mod 13
h(24)=77 mod 13
h(24)=**12**

0	
1	3
2	
3	
4	
5	13
6	
7	
8	
9	
10	6
11	
12	24

X_5 = 18, f(18) = 3*18+5=59.
h(18) = (59 + 0 + 0^2) mod 13
h(18)=59 mod 13
h(18)= **7**

0	
1	3
2	
3	
4	
5	13
6	
7	18
8	
9	
10	6
11	
12	24

X_6 = 2, f(2) = 3*2+5=11.
h(2) = (11 + 0 + 0^2) mod 13
h(2)=11 mod 13
h(2)=**11**

0	
1	3
2	
3	
4	
5	13
6	
7	18
8	
9	
10	6
11	2
12	24

X_7 = 5, f(5) = 3*5+5=20
h(5) = (20 + 0 + 0^2) mod 13
h(5)=20 mod 13
h(5)=**7 //collision, i++**

h_1(5) = (20 + 1 + 1^2) mod 13
h_1(5)=22 mod 13
h_1(5)= **9**

0	
1	3
2	
3	
4	
5	13
6	
7	18
8	
9	5
10	6
11	2
12	24

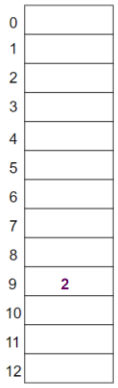
Double Hashing

Formula: $h(x) = (h_1(x) + i * h_2(x)) \bmod M$ $M > N$, nr i thjeshtë
$h_1(x) = (f(x) \bmod M)$ $i = 0,1,2\ldots$
$h_2(x) = M - (f(x) \bmod M)$

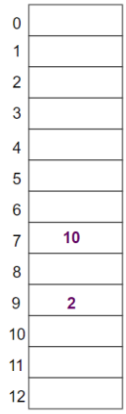
Detyra 1: Të bëhet hash vargu i dhënë A = {2,10,3,6,18,13,11,5}, nëse është dhënë f(x) = 3x + 3

N = 8, M = 13

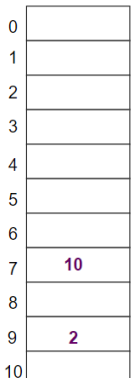
$X_1 = 2$, f (2) = 6+3= 9
$h_1(2) = 9 \bmod 13 = 9$
$h_2(2) = 13 - (9 \bmod 13) = 13 - 9 = 4$
$h(2) = (9 + 0 * 4) \bmod 13 = 9 \bmod 13 =$ **9**



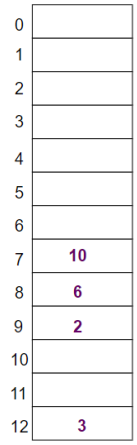
$X_2 = 10$, f (10) = 30+3= 33
$h_1(10) = 33 \bmod 13 = 7$
$h_2(10) = 13 - (33 \bmod 13) = 13 - 7 = 6$
$h (10) = (7 + 0 * 6) \bmod 13 = 7 \bmod 13 =$ **7**



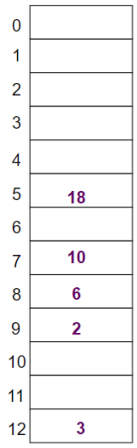
X_3 = 3, f (3) = 9+3= 12
$h_1(3)$ = 12mod13 = 12
$h_2(3)$ = 13 - (12mod13) = 13 - 12 = 1
h (3) = (12 + 0 * 1) mod13 = 12mod13 = **12**



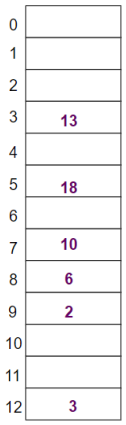
X_4 = 6, f (6) = 18+3= 21
$h_1(6)$ = 21mod13 = 8
$h_2(6)$ = 13 - (21mod13) = 13 - 8 = 5
h (6) = (8 + 0 * 5) mod13 = 8mod13 = **8**



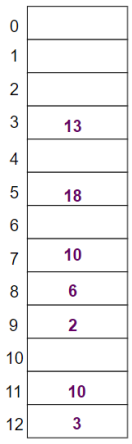
X_5 = 18, f (18) = 54+3= 57
$h_1(18)$ = 57mod13 = 5
$h_2(18)$ = 13 - (51mod13) = 13-5= 8
h (18) = (5 + 0 * 8) mod13 = 5mod13 = **5**



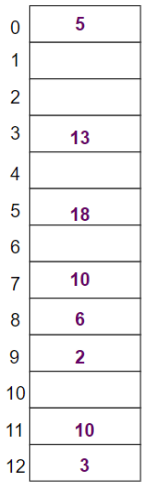
X_6 = 13, f (13) = 39+3= 42
$h_1(13)$ = 42mod13 = 3
$h_2(13)$ = 13 - (42mod13) = 13-3=10
h (13) = (3 + 0 * 10) mod13 = 3mod13 = **3**



X_7 = 11, f (11) = 33+3= 36
$h_1(11)$ = 36mod13 = 10
$h_2(11)$ = 13 - (36mod13) = 13-10=3
h (11) = (10 + 0 * 3) mod13 = 10mod13 = **10**



X_8 = 5, f (5) = 15+3= 18
$h_1(5)$ = 18mod13 = 5
$h_2(5)$ = 13 - (18mod13) = 13-5=8
h (5) = (5 + 0 * 8) mod13 = 5mod13 = 5 // **collision, i++**
h (5) = (5 + 1 * 8) mod13 = 13mod13 = **0**



Seperate Chaining (Closed Addressing)

Formula: h(x) = f(x) mod M M < N , nr i thjeshtë

→ Në rastet kur në tabelë në asnjë nga index-at nuk vendosen të paktën **3 keys**, duhet ta bëjmë resize tabelën (table size zvoglohet) → **M / 2**
→ Në rastet kur në tabelë të paktën në njërin nga index-at janë vendosur më shumë se **5 keys**, duhet ta bëjmë resize tabelën (table size rritet) → **M * 2**

Detyra 1: Të bëhet hash vargu i dhënë: A={14, 10, 7, 2, 5, 6, 9} nëse është dhënë
f(x) = 2x + 5

N = 7, M = 5

X_1 = 14, f (14) = 2*14 +5= 33
h(14) = 33 mod5 = **3**

X2 =10 , f (10) = 2*10 +5= 25
h(14) = 25 mod5 = **0**

X3 =7 , f (7) = 2*7 +5=19

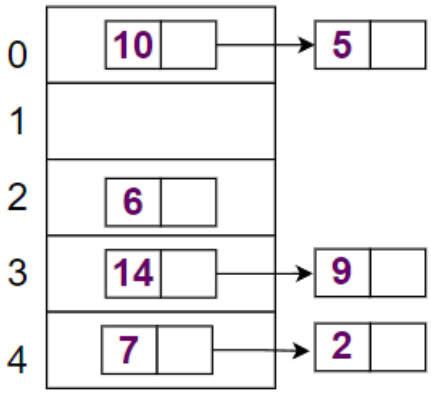
h(7) = 19mod5 =**4**

X4 =2 , f (2) = 2*2 +5=9
h(2) = 9mod5 =**4**

X5 =5 , f (5) = 2*5 +5=15
h(2) = 15mod5 =**0**

X6 =6 , f (6) = 2*6 +5=17
h(6) = 17mod5 =**2**

X7 =9 , f (9) = 2*9+5=23
h(9) = 23mod5 =**3**



Pasi që në asnjë nga index-at nuk janë vendosur të paktën **3 keys,** duhet ta zvoglojmë table size.
Marrim **M = 3**

X_1 = 14, f (14) = 2*14 +5= 33
h(14) = 33 mod3 = **0**

X2 =10 , f (10) = 2*10 +5= 25
h(14) = 25 mod3 = **1**

X3 =7 , f (7) = 2*7 +5=19
h(7) = 19mod3 =**1**

X4 =2 , f (2) = 2*2 +5=9
h(2) = 9mod3 =**0**

X5 =5 , f (5) = 2*5 +5=15
h(2) = 15mod3 =**0**

X6 =6 , f (6) = 2*6 +5=17
h(6) = 17mod3 =**2**

X7 =9 , f (9) = 2*9+5=23
h(9) = 23mod3 =**2**

